

2.1 The BIOS: What is it?

The BIOS (Basic Input/Output System) is the software code that first runs when the PC powers on. It contains all the information necessary to initialise nearly all the hardware components of the PC. Normally, when you switch on the PC, the BIOS performs a Power on Self Test, or POST as it is called. This is a series of diagnostic tests on the RAM and other hardware. It also initialises all the hardware devices such as the hard disk, memory, video and other hardware, identifies and reserves memory addresses for all the IRQs and ports available on the motherboard, and calls a small operating system program known as the boot loader. The boot loader, using the BIOS information amongst other things, starts calling the programs that will load the OS. And finally, the OS uses the BIOS information to take control over the hardware devices.

Motherboard manufacturers use the BIOS to define settings for the various hardware components including the hard disk, RAM, CD-ROMs, I/O ports etc. These are generally set at the factory and are what is called the Factory Settings or the BIOS Setup Default Settings.

EFI—The End of BIOS?

The BIOS as we know it is well over 20 years old, and has changed little since it was first introduced. In early 2000, Intel indicated that this was about to change with the release of the first version of the Extensible Firmware Interface (EFI). The EFI is a tiny operating system that has its own basic networking, graphics, keyboard and storage handling software. It will allow hardware vendors to create OS-independent device drivers. This means that whatever the operating system, your hardware device would work, at least theoretically, without you having to install specific drivers. Bye bye to Device Manager hell? It's still too early to say. Windows Vista will be the first Windows version to offer native EFI support, and hardware vendors have been reluctant to include EFI due to lack of mainstream (read Windows!) demand.